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Approximation Algorithms and Approximation Ratios: Introduction

Jobshop Minimum Makespan Scheduling

Vertex Cover

Maximum Satisfiability Problem

✔

Video: Approximation Algorithms for the Maximum Satisfiability Problem

28 min

✔

Lab: Interactive Notes on Maximum Satisfiability Approximation

1h

❗

Quiz: Max-SAT Approximation

Submitted

Problem Set # 3

Max-SAT Approximation

Review Learning Objectives

✔ Submit your assignment

Due Feb 25, 11:59 PM IST

✔ Receive grade

To Pass 80% or higher

✔ Congratulations! You passed!

Grade received 100%

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The problems below concern the following instance for the MAX-SAT problem involving propositional variables x_1, x_2, x_3, x_4 .
Note that $\vee$ stands for Or, $\overline{x_j}$ represents the negation of x_j

Consider the MAXSAT instance involving the following clauses.

$$\begin{array}{l} x_1 \vee x_2 \vee \overline{x_4} \\ \overline{x_2} \vee x_3 \vee x_4 \\ x_1 \vee \overline{x_2} \vee x_3 \\ x_2 \vee \overline{x_3} \vee \overline{x_4} \end{array}$$

Try again

Your grade

100%

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1. What is the expected number of clauses that a randomly chosen truth assignment where each variable is independently assigned true/false with probability $\frac{1}{2}$.

3 / 3 points

3.5

✔ Correct

2. Using just the information from the previous problem select which fact you can conclude.

2 / 2 points

- ☒ There is a truth assignment that satisfies all the clauses.
- ☐ None of the other options provided here are valid.
- ☐ We cannot do better than satisfy three out of four clauses for the problem above.

✔ Correct

True: the average is 3.5 So there has to be at least one assignment that does as well or better than the average. However, number of clauses satisfied is a whole number. Therefore, at least one assignment satisfies all four clauses.

3. Suppose we set $x_1 = true$, what is the expected number of clauses we can satisfy in the resulting formula?

2 / 2 points

3.75

✔ Correct

Setting $x_1 = true$ satisfies two clauses outright. The other two clauses have three variables each and thus expected number satisfied is $2 * 7/8 = 1.75$. Overall, we get an expected number 3.75 satisfied clauses.

4. Suppose we set $x_1 = false$, what is the expected number of clauses we can satisfy in the resulting formula?

2 / 2 points

3.25

✔ Correct

Setting $x_1 = false$ makes the first and third clause have two remaining literals each. The remaining two clauses (second and fourth) have three literals each. This yields expected number of clauses satisfied $= 2 \times 3/4 + 2 \times 7/8 = 2 \times 13/8 = 13/4 = 3.25$

5. What truth value will our approximation algorithm for MAXSAT assign to x_1 ?

1 / 1 point

- ☒ True
- ☐ False

✔ Correct

Correct. Setting $x_1 = true$ yields larger expected number of satisfied clauses.

